

Comparative Analysis of Deep Learning Architectures for Smartphone Sensor-Based Human Activity Recognition

**Synopsis report on
Deep Learning Project
[ICT-4442]**

Submitted By

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PART A: SYNOPSIS

Team No. and Names (with Registration Numbers)

Team No.	Member Name	Registration No.
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Title of the Project

Comparative Analysis of Deep Learning Architectures for Smartphone Sensor-Based Human Activity Recognition.

Brief Description of the Project

Human Activity Recognition (HAR) is the task of automatically identifying a person's physical activity — such as walking, climbing stairs, sitting, or standing — from motion signals captured by the accelerometer and gyroscope embedded in an ordinary smartphone. It underpins many real-world systems, including fitness and step-tracking apps, fall detection and remote monitoring for the elderly, rehabilitation tracking, and context-aware mobile services. Because inertial signals are time-series data, no single neural architecture is obviously best: convolutional, recurrent, and attention-based models each capture temporal structure differently, which makes HAR an ideal setting for a controlled comparison of deep-learning families.

This project implements and compares four models spanning distinct architecture families on a single common evaluation protocol: (1) a Multi-Layer Perceptron (MLP) baseline trained on the 561 hand-crafted time- and frequency-domain features; (2) a 1D Convolutional Neural Network (CNN) that learns local motion patterns directly from the raw nine-channel inertial signals; (3) an LSTM/BiLSTM recurrent network that models sequential temporal dependencies; and (4) a Transformer encoder that uses self-attention to capture long-range dependencies in parallel. Each model is owned by one team member, so every member's contribution is independently demonstrable and defensible in the viva.

The primary dataset is the UCI “Human Activity Recognition Using Smartphones” dataset (Anguita et al., 2013), which provides both the engineered feature vectors and the raw windowed signals, allowing the feature-based baseline and the raw-signal deep models to be compared fairly on the same subjects and splits. All four models will be trained and tested on an identical train/validation/test partition and evaluated using the same metrics — accuracy and macro-averaged precision, recall, and F1-score — with confusion matrices for per-class error analysis. Results will be consolidated into a single comparison table, and the report will explain *why* performance differs across architectures (for example, the well-known confusion between sitting and standing, and between the three walking variants). The expected outcome is a clear, evidence-backed account of the accuracy–complexity trade-offs among MLP, CNN, LSTM, and Transformer models for on-device HAR, giving practical guidance on which family suits resource-constrained wearable deployment.

Dataset(s) Identified

Primary dataset — UCI HAR (Human Activity Recognition Using Smartphones), Anguita et al., 2013.

- Source: UCI Machine Learning Repository (Dataset ID 240); DOI 10.24432/C54S4K. Licensed under CC BY 4.0 (free for academic use).
- Subjects & device: 30 volunteers aged 19–48, each wearing a waist-mounted Samsung Galaxy S II; signals from the built-in tri-axial accelerometer and gyroscope.
- Activities (6 classes): walking, walking upstairs, walking downstairs, sitting, standing, and lying.
- Sampling & windowing: 50 Hz sampling; fixed-width sliding windows of 2.56 s (128 readings) with 50% overlap.
- Size & split: 10,299 instances—7,352 for training and 2,947 for testing (subject-independent split).
- Two representations provided: (a) 561 engineered time/frequency features per window (used by the MLP baseline); (b) 9 raw inertial signal channels of 128 timesteps each—body acceleration, total acceleration, and angular velocity in x/y/z (used by the CNN, LSTM, and Transformer).

Secondary / optional dataset for cross-validation of findings: WISDM (Weiss, 2019) — accelerometer data from 36 subjects across similar activities — kept as a stretch goal to test whether the ranking of architectures generalises beyond a single dataset.

Proposed Models for Comparison

#	Model (Architecture Family)	Input Representation	Role in the Comparison
1	MLP — Multi-Layer Perceptron (classical / feed-forward baseline)	561 hand-crafted time & frequency-domain features	Reference baseline; tests how far engineered features alone can go without temporal modelling.
2	1D CNN — Convolutional Neural Network	Raw 9-channel inertial signals (128 timesteps)	Learns local motion patterns automatically from raw signals; no manual feature engineering.
3	LSTM / BiLSTM — Recurrent Neural Network	Raw 9-channel inertial signals (128 timesteps)	Models sequential temporal dependencies across the window.
4	Transformer Encoder — Attention-based model	Raw 9-channel inertial signals (128 timesteps)	Uses self-attention to capture long-range dependencies in parallel; current state of the art.

Model-to-Member Assignment

Member	Model / Task Owned	Also Responsible For
Daasharathi Darahaas Yandam	MLP baseline (feature-based)	Feature-set preparation & train/val/test split definition
Akshaj Vinod Chandak	1D CNN	Raw-signal loading & windowing pipeline
Nandan Reddy Maram	LSTM / BiLSTM	Evaluation harness & common metrics module
K V Ananth Karantha	Transformer encoder	Comparison table, error analysis & report integration

Shared work — preprocessing pipeline, common evaluation protocol, and repository maintenance — is distributed as shown so that every member has an independently verifiable model plus a distinct supporting responsibility. All members commit to a shared GitHub repository with per-member commit history.

Existing State-of-the-Art Method(s) Related to the Proposed Project

- Feature + classical ML: The original UCI HAR work used 561 engineered features with a multiclass SVM, reaching roughly 90–96% accuracy — the benchmark our MLP baseline is measured against [1].
- CNN on raw signals: Ronao and Cho showed deep 1D CNNs on raw smartphone signals reach ~95% accuracy without hand-crafted features, establishing CNNs as a strong HAR approach [2].
- Hybrid CNN-LSTM: The DeepConvLSTM architecture of Ordóñez and Roggen combines convolution and recurrence for multimodal wearable HAR and remains a widely cited state-of-the-art design [3].
- Systematic family comparison: Hammerla et al. benchmarked deep, convolutional, and recurrent models on wearable HAR and gave guidance on when each family wins [4].
- Attention / Transformer: Recent Transformer-based HAR models (e.g., Dirgová Luptáková et al., 2022) match or exceed CNN-LSTM performance using self-attention, representing the current state of the art [5], [6].

Objectives

O1. A validated, like-for-like comparison of MLP, CNN, LSTM, and Transformer architectures for smartphone-based HAR under one common evaluation protocol and metric set.

O2. A characterisation of the accuracy–complexity trade-off across the four families — including the value of raw-signal learning versus hand-crafted features — to inform model selection for resource-constrained wearable deployment.

O3. An error analysis identifying which activities are hardest to distinguish and explaining why performance differs across architectures.

Importance / Motivation of the Proposed Project

Smartphone- and wearable-based HAR has direct real-world impact: it powers fitness and wellness tracking, enables fall detection and continuous monitoring for elderly and post-operative patients, and supports rehabilitation and context-aware applications — all using sensors people already carry, which is cheaper and more privacy-preserving than camera-based monitoring. From a learning standpoint, HAR is an ideal vehicle for this course because its inertial time-series lets classical, convolutional, recurrent, and attention-based models genuinely compete on the same task, so the comparison is meaningful rather than decided in advance. The project therefore delivers both a practically useful result — guidance on which architecture best balances accuracy and efficiency for on-device recognition — and a rigorous, hands-on comparison of the core deep-learning families covered in the syllabus.

Preliminary Reference List

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- [4] N. Y. Hammerla, S. Halloran, and T. Plötz, “Deep, Convolutional, and Recurrent Models for Human Activity Recognition using Wearables,” in Proc. 25th Int. Joint Conf. on Artificial Intelligence (IJCAI), 2016, pp. 1533–1540.
- [5] I. Dirgová Luptáková, M. Kubovčík, and J. Pospíchal, “Wearable Sensor-Based Human Activity Recognition with Transformer Model,” *Sensors*, vol. 22, no. 5, art. 1911, 2022.
- [6] A. Vaswani et al., “Attention Is All You Need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [7] O. D. Lara and M. A. Labrador, “A Survey on Human Activity Recognition using Wearable Sensors,” *IEEE Communications Surveys & Tutorials*, vol. 15, no. 3, pp. 1192–1209, 2013.
- [8] A. Bulling, U. Blanke, and B. Schiele, “A Tutorial on Human Activity Recognition Using Body-Worn Inertial Sensors,” *ACM Computing Surveys*, vol. 46, no. 3, pp. 1–33, 2014.

Work Plan

Sl. No.	Component / Work Element / Milestone	Expected Completion
1	Team formation, problem finalisation & synopsis submission	25 Aug 2026
2	Dataset acquisition, exploratory analysis & preprocessing pipeline (windowing, normalisation, splits)	05 Sep 2026
3	Implement MLP baseline and 1D CNN with preliminary (unoptimised) results	18 Sep 2026
4	Literature review table (8–10 papers) + Interim report submission	25 Sep 2026
5	Implement LSTM/BiLSTM and Transformer models	12 Oct 2026
6	Hyperparameter tuning, common-protocol evaluation & comparison table	20 Oct 2026
7	Error analysis, final report writing & presentation preparation	25 Oct 2026
8	Final presentation (viva) and report + code repository submission	26–31 Oct 2026